

Assignment1

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Tasks for the Assignment

- Generate a random sample of size 100 from the standard normal distribution and plot the empirical quantile function for your sample.
- Find the true 35th and 55th quantiles of the standard normal distribution.
- Compute the 35th and 55th empirical quantiles from your data. Are they close to the true/theoretical ones?
- What would you do to get empirical quantiles closer to the theoretical ones?

Assignment discussion:

The empirical quantile function is closely related to the empirical cumulative distribution function (ECDF). In fact, it can be viewed as a generalized inverse of the ECDF.

Suppose we have a random sample

$$X_1, X_2, \dots, X_n.$$

After sorting the observations, we obtain the order statistics

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

The empirical cumulative distribution function is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x),$$

where $I(\cdot)$ is the indicator function. The ECDF gives the proportion of observations that are less than or equal to a particular value x .

Thus, the ECDF maps a data value to a probability:

$$x \longmapsto F_n(x).$$

The empirical quantile function reverses this relationship. Given a probability p , it returns the smallest observed value whose empirical cumulative probability is at least p .

Formally,

$$Q_n(p) = \inf \{x : F_n(x) \geq p\}, \quad 0 < p \leq 1.$$

Therefore, the empirical quantile function maps

$$p \mapsto Q_n(p).$$

For a sample of size n , the ECDF increases by increments of

$$\frac{1}{n}.$$

Consequently, the empirical quantile function can be described using the order statistics. If

$$\frac{k-1}{n} < p \leq \frac{k}{n},$$

then

$$Q_n(p) = X_{(k)}.$$

Equivalently,

$$Q_n(p) = X_{(\lceil np \rceil)},$$

where $\lceil np \rceil$ denotes the smallest integer greater than or equal to np .

Example

Suppose the sorted sample is

$$2, 4, 5, 9, 12.$$

Here,

$$n = 5,$$

so each observation corresponds to a probability increment of

$$\frac{1}{5} = 0.2.$$

The empirical quantile function is therefore

$$Q_n(p) = \begin{cases} 2, & 0 < p \leq 0.2, \\ 4, & 0.2 < p \leq 0.4, \\ 5, & 0.4 < p \leq 0.6, \\ 9, & 0.6 < p \leq 0.8, \\ 12, & 0.8 < p \leq 1. \end{cases}$$

For example,

$$Q_n(0.35) = 4$$

because the second observation is the first value for which the ECDF reaches or exceeds 0.35. Similarly,

$$Q_n(0.75) = 9.$$

Relationship to the ECDF

The ECDF and empirical quantile function contain essentially the same information, but the axes are reversed.

For the ECDF,

$$x \rightarrow F_n(x),$$

so the horizontal axis contains observed data values and the vertical axis contains cumulative probabilities.

For the empirical quantile function,

$$p \rightarrow Q_n(p),$$

so the horizontal axis contains probabilities and the vertical axis contains observed data values.

Both functions are step functions because they are constructed from a finite sample.

Plotting the empirical quantile function in R

The empirical quantile function can be plotted by sorting the observations and associating them with the probability levels

$$\frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n}.$$

For example:

```
x_sorted <- sort(x)
n <- length(x)

plot(
  (1:n) / n,
  x_sorted,
  type = "s",
  xlab = "p",
  ylab = "Empirical Quantile",
  main = "Empirical Quantile Function"
)
```

The vector

```
(1:n) / n
```

constructs the empirical probability levels

$$\frac{1}{n}, \frac{2}{n}, \dots, 1,$$

while

```
sort(x)
```

produces

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}.$$

Thus each plotted point corresponds conceptually to

$$\left(\frac{k}{n}, X_{(k)} \right).$$

The step plot represents the empirical quantile function obtained from the observed sample.

Empirical versus theoretical quantiles

If the sample comes from a distribution with cumulative distribution function $F(x)$, then the theoretical quantile function is

$$Q(p) = F^{-1}(p).$$

For example, for a standard normal distribution, the theoretical quantile function is

$$Q(p) = \Phi^{-1}(p).$$

In R, this is calculated using

```
qnorm(p)
```

while empirical quantiles are obtained from the observed sample.

As the sample size increases, the empirical quantile function generally becomes a better approximation to the population quantile function.

```
set.seed(123)

x <- rnorm(100)
n <- length(x)

plot(
  (1:n) / n,
  sort(x),
  type = "s",
  xlab = "p",
  ylab = "Empirical Quantile",
```

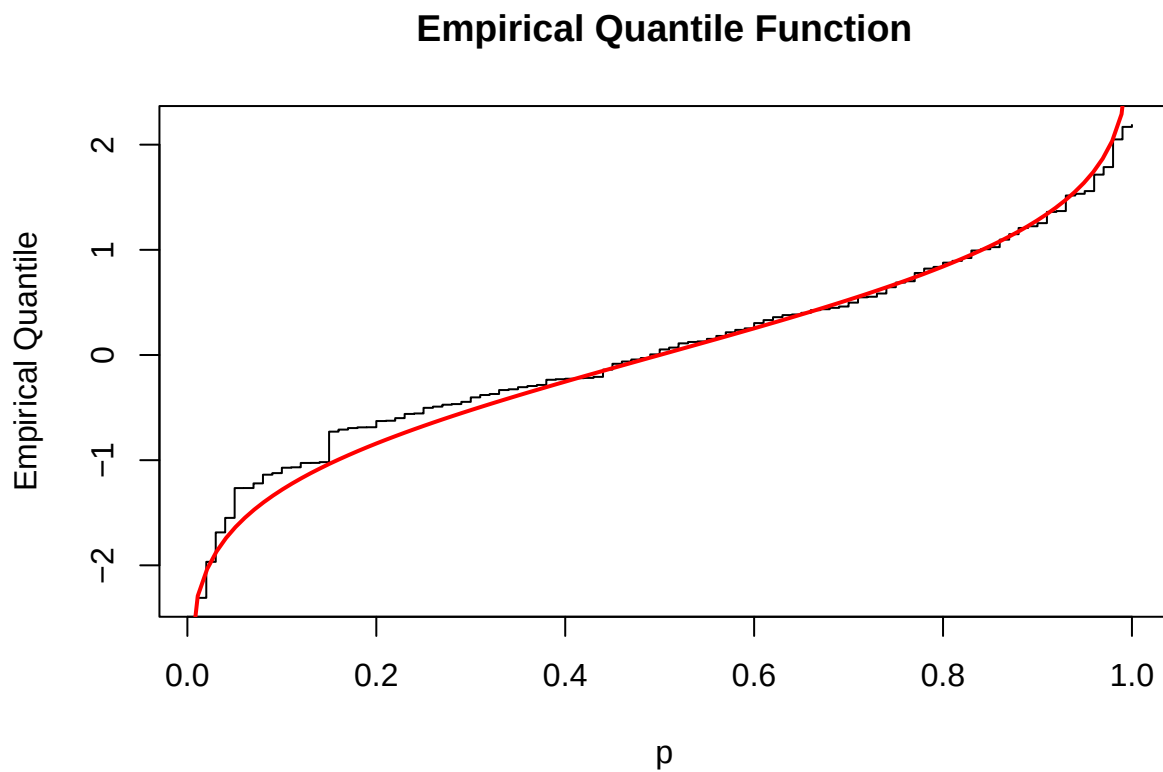
```

    main = "Empirical Quantile Function"
)

# overlay the Theoretical Normal Quantile function

curve(
  qnorm(x),
  from = 0.001,
  to = 0.999,
  add = TRUE,
  col = "red",
  lwd = 2
)

```



```

x_sorted <- sort(x)
#cat("vector length is", n, "\n")
q_55 <- x_sorted[ceiling(0.55 * n)] #return the 55th element of the sorted vector
cat("Empirical Q_55 = ", q_55, "\n")

```

```
## Empirical Q_55 = 0.1813035
```

```
cat("Analytic Q_55", qnorm(.55), "\n\n")
```

```
## Analytic Q_55 0.1256613
```

```
q_35 <- x_sorted[ceiling(0.35 * n)] # return the 35th element of the sorted vector.  
cat("Emperical Q_35 = ", q_35, "\n")
```

```
## Emperical Q_35 = -0.3059627
```

```
cat("Analytic Q_35", qnorm(.35), "\n")
```

```
## Analytic Q_35 -0.3853205
```

```
set.seed(123)
```

```
x <- rnorm(1000)
```

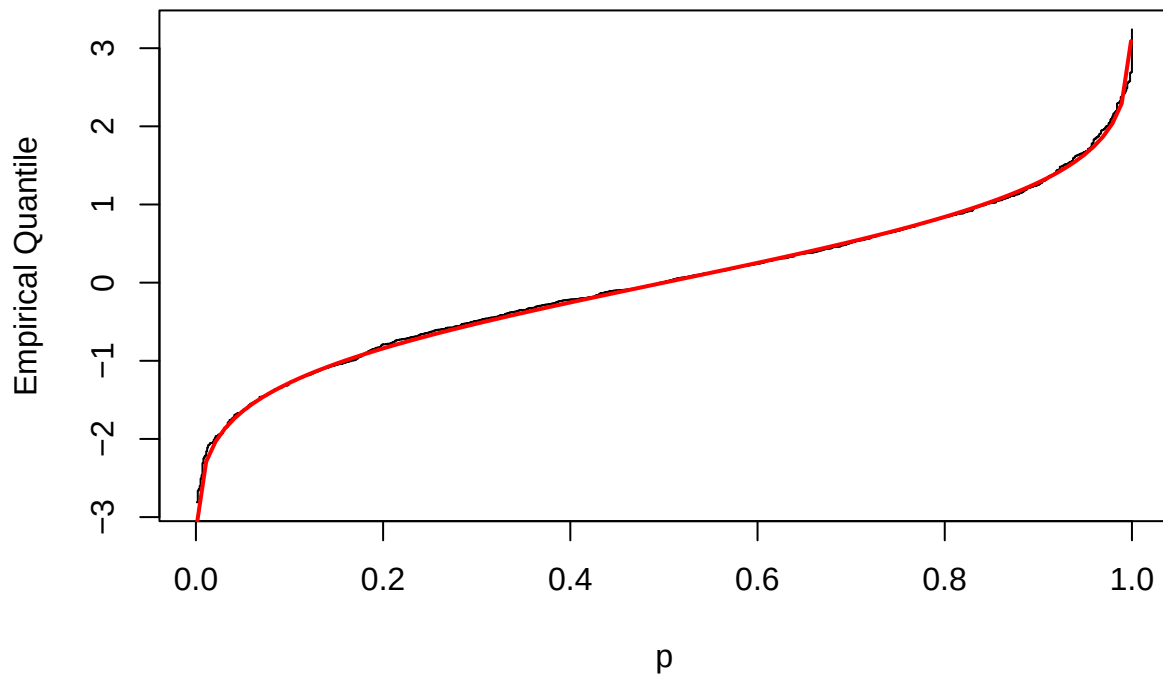
```
n <- length(x)
```

```
plot(  
  (1:n) / n,  
  sort(x),  
  type = "s",  
  xlab = "p",  
  ylab = "Empirical Quantile",  
  main = "Empirical Quantile Function"  
)
```

```
# overlay the Theoritical Normal Quantile function
```

```
curve(  
  qnorm(x),  
  from = 0.001,  
  to = 0.999,  
  add = TRUE,  
  col = "red",  
  lwd = 2  
)
```

Empirical Quantile Function



```
x_sorted <- sort(x)
# cat("vector length is", n, "\n")
q_55 <- x_sorted[ceiling(0.55 * n)] # return the 550th element of the sorted vector
cat("Emperical Q_55 = ", q_55, "\n")
```

```
## Emperical Q_55 = 0.1263159
```

```
cat("Analytic Q_55", qnorm(.55), "\n\n")
```

```
## Analytic Q_55 0.1256613
```

```
q_35 <- x_sorted[ceiling(0.35 * n)] # return the 350th element of the sorted vector
cat("Emperical Q_35 = ", q_35, "\n")
```

```
## Emperical Q_35 = -0.3496504
```

```
cat("Analytic Q_35", qnorm(.35), "\n")
```

```
## Analytic Q_35 -0.3853205
```